

Load Balancing in Docker

Name: Pooja Prakash Belgaumkar

1. First, we prepare the application code required for load balancing.

```

• MobaXterm Personal Edition v25.4 •
(SSH client, X server and network tools)

► SSH session to ec2-user@51.20.79.34
• Direct SSH : ✓
• SSH compression : ✓
• SSH-browser : ✓
• X11-forwarding : ✗ (disabled or not supported by server)

► For more info, ctrl+click on help or visit our website.

```



```

Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

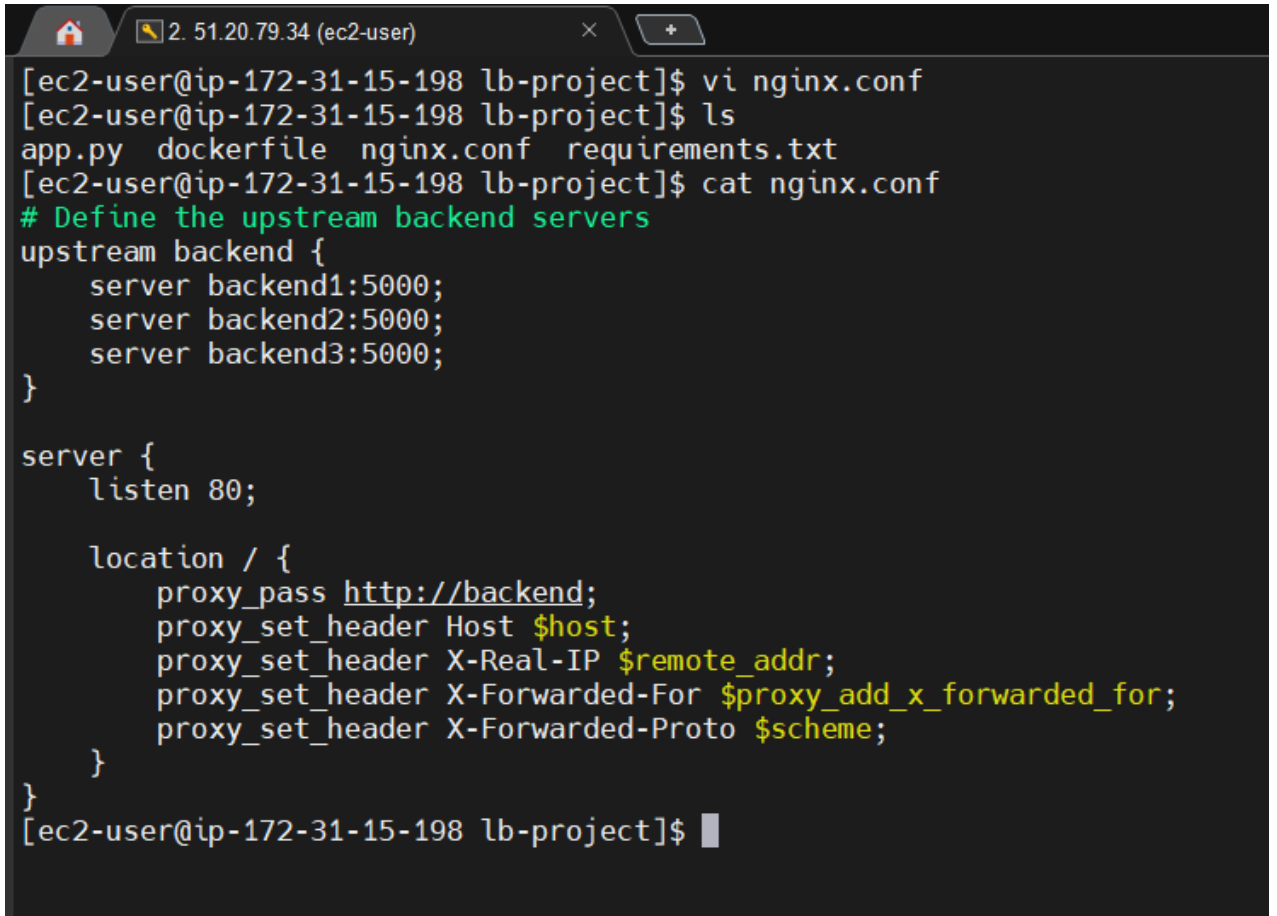
Last login: Sat Jan 24 20:35:06 2026 from 171.61.183.254
[ec2-user@ip-172-31-15-198 ~]$ mkdir lb-project
[ec2-user@ip-172-31-15-198 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-15-198 ~]$ cd lb-project
[ec2-user@ip-172-31-15-198 lb-project]$ vi app.py
[ec2-user@ip-172-31-15-198 lb-project]$

```

2. Since our application is in Python, we add the requirements.txt file and write a Dockerfile to create Docker images.

```
[ec2-user@ip-172-31-15-198 ~]$ mkdir lb-project
[ec2-user@ip-172-31-15-198 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-15-198 ~]$ cd lb-project
[ec2-user@ip-172-31-15-198 lb-project]$ vi app.py
[ec2-user@ip-172-31-15-198 lb-project]$ vi requirements.txt
[ec2-user@ip-172-31-15-198 lb-project]$ vi dockerfile
[ec2-user@ip-172-31-15-198 lb-project]$ cat dockerfile
from python:3.9
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python", "app.py"]
[ec2-user@ip-172-31-15-198 lb-project]$
```

3. Here, Nginx acts as a load balancer in Docker and serves traffic among three backend servers.

A terminal window with a dark background and light-colored text. The window title bar shows a home icon, a network icon, and the text '2. 51.20.79.34 (ec2-user)'. The terminal content shows a series of commands and their outputs. The user enters 'vi nginx.conf', then 'ls', which lists 'app.py', 'dockerfile', 'nginx.conf', and 'requirements.txt'. Then the user enters 'cat nginx.conf', which displays the Nginx configuration. The configuration defines an upstream named 'backend' with three servers: 'backend1:5000', 'backend2:5000', and 'backend3:5000'. It also defines a server listening on port 80, with a location '/' that proxies requests to 'http://backend' and sets various headers like 'Host', 'X-Real-IP', 'X-Forwarded-For', and 'X-Forwarded-Proto'. The prompt returns to the shell after the configuration is displayed.

```
[ec2-user@ip-172-31-15-198 lb-project]$ vi nginx.conf
[ec2-user@ip-172-31-15-198 lb-project]$ ls
app.py  dockerfile  nginx.conf  requirements.txt
[ec2-user@ip-172-31-15-198 lb-project]$ cat nginx.conf
# Define the upstream backend servers
upstream backend {
    server backend1:5000;
    server backend2:5000;
    server backend3:5000;
}

server {
    listen 80;

    location / {
        proxy_pass http://backend;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
[ec2-user@ip-172-31-15-198 lb-project]$
```

4. Then, we prepare the docker-compose.yml file and add it to the project directory.

```
services:
  nginx:
    image: nginx:latest
    container_name: nginx-lb
    ports:
      - "80:80"
    volumes:
      - ./nginx.conf:/etc/nginx/conf.d/default.conf
    depends_on:
      - backend1
      - backend2
      - backend3

  backend1:
    build: .
    container_name: backend1
    expose:
      - "5000"

  backend2:
    build: .
    container_name: backend2
    expose:
      - "5000"

  backend3:
    build: .
    container_name: backend3
    expose:
      - "5000"
```

5. Next, we build the Docker images and start the containers using Docker Compose.

```
[ec2-user@ip-172-31-15-198 lb-project]$ vi docker-compose.yml
[ec2-user@ip-172-31-15-198 lb-project]$ docker compose up -d --build
[+] Running 8/8
 ✓ nginx Pulled 2.5s
   ✓ 119d43eec815 Already exists 0.0s
   ✓ 700146c8ad64 Already exists 0.0s
   ✓ d989100b8a84 Already exists 0.0s
   ✓ 500799c30424 Already exists 0.0s
   ✓ 10b68cfefee1 Already exists 0.0s
   ✓ 57f0dd1befe2 Already exists 0.0s
   ✓ eaf8753feae0 Already exists 0.0s
[+] Building 22.4s (13/21) docker:default
=> => transferring dockerfile: 233B 0.0s
=> [backend3 internal] load build definition from dockerfile 0.1s
=> => transferring dockerfile: 233B 0.0s
=> [backend3 internal] load metadata for docker.io/library/python:3.9 1.6s
=> [backend2 internal] load .dockerignore 0.1s
```

```
[+] Running 5/5
✓ Network lb-project_default Created 0.2s
✓ Container backend1 Started 1.1s
✓ Container backend3 Started 0.9s
✓ Container backend2 Started 0.9s
✓ Container nginx-lb Started 1.7s
[ec2-user@ip-172-31-15-198 lb-project]$ docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
lb-project-backend3  latest      1381f7de5c9e     About a minute ago 1.1GB
lb-project-backend2  latest      2708e32a09fe     About a minute ago 1.1GB
lb-project-backend1  latest      3426379be85a     About a minute ago 1.1GB
kastro-ecom-dc-order-service latest      560590c207d3     2 days ago        243MB
kastro-ecom-dc-product-service latest      8e01b3122fa7     2 days ago        243MB
kastro-ecom-dc-user-service latest      7f4151128d8b     2 days ago        243MB
kastro-ecom-dc-main-service latest      9ee86f94ffae     2 days ago        243MB
kastro-ecom-dc-cart-service latest      dce7f9d2d8cd     2 days ago        243MB
nginx                latest      4af177a024eb     13 days ago       161MB
[ec2-user@ip-172-31-15-198 lb-project]$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS
NAMES
448c25fb75a1   nginx:latest   "/docker-entrypoint..." 2 minutes ago Up 2 minu
tes  0.0.0.0:80->80/tcp, :::80->80/tcp nginx-lb
6bffa0067473   lb-project-backend1 "python app.py"         2 minutes ago Up 2 minu
tes  5000/tcp backend1
20ea9e67de42   lb-project-backend2 "python app.py"         2 minutes ago Up 2 minu
tes  5000/tcp backend2
3f279a41fd67   lb-project-backend3 "python app.py"         2 minutes ago Up 2 minu
tes  5000/tcp backend3
[ec2-user@ip-172-31-15-198 lb-project]$
```

6. Finally, we access the application in the browser, where the container hostname changes on every page refresh, showing that load balancing is working.
7. Nginx distributes requests among multiple Docker containers, and Docker Compose is used to manage and run them together.

← → ↻ ⚠ Not secure 51.20.79.34 ☆ 🌧 📦 📶

container_response

Response from Docker container instance

HOSTNAME 6a855695df6a	CONTAINER ID 6a855695df6a
CURRENT TIME 3:18:48 AM	PYTHON VERSION 3.9.25
PLATFORM Linux 6.1.159-182.297.amzn2023.x86_64	STATUS ● Operational

Deployed with ❤ by Kastro Kiran V

Request served from: 6a855695df6a

← → ↻ ⚠ Not secure 51.20.79.34 ☆ 🌧 📦 📶

container_response

Response from Docker container instance

HOSTNAME e3c027bc36b0	CONTAINER ID e3c027bc36b0
CURRENT TIME 3:19:39 AM	PYTHON VERSION 3.9.25
PLATFORM Linux 6.1.159-182.297.amzn2023.x86_64	STATUS ● Operational

Deployed with ❤ by Kastro Kiran V

Request served from: e3c027bc36b0